standard 0.707 (-3dB) used in analog electronics and other digital filters? This is because the windowed-sinc's frequency response is *symmetrical* between the passband and the stopband. For instance, the Hamming window results in a passband ripple of 0.2%, and an *identical* stopband attenuation (i.e., ripple in the stopband) of 0.2%. Other filters do not show this symmetry, and therefore have no advantage in using the one-half amplitude point to mark the cutoff frequency. As shown later in this chapter, this symmetry makes the windowed-sinc ideal for *spectral inversion*.

After f_C and M have been selected, the filter kernel is calculated from the relation:

$$h[i] = K \frac{\sin(2\pi f_C (i - M/2))}{i - M/2} \left[0.42 - 0.5\cos\left(\frac{2\pi i}{M}\right) + 0.08\cos\left(\frac{4\pi i}{M}\right) \right]$$

EQUATION 16-4
The windowed-sinc filter kernel. The cutoff frequency, f_C, is expressed as a fraction of the sampling rate, a value between 0 and 0.5. The length of the filter kernel is determined by M, which must be an even integer. The sample number i, is an integer that runs from 0 to M, resulting in $M+1$ total points in the filter kernel. The constant, K, is chosen to provide unity gain at zero frequency. To avoid a divide-by-zero error, for $i = M/2$, use $h[i] = 2\pi f_C K$.

Don't be intimidated by this equation! Based on the previous discussion, you should be able to identify three components: the *sinc function*, the *M/2 shift*, and the *Blackman window*. For the filter to have unity gain at DC, the constant K must be chosen such that the sum of all the samples is equal to one. In practice, ignore K during the calculation of the filter kernel, and then *normalize* all of the samples as needed. The program listed in Table 16-1 shows how this is done. Also notice how the calculation is handled at the center of the sinc, $i = M/2$, which involves a division by zero.

This equation may be long, but it is easy to use; simply type it into your computer program and forget it. Let the computer handle the calculations. If you find yourself trying to evaluate this equation by hand, you are doing something very very wrong.

Let's be specific about where the filter kernel described by Eq. 16-4 is located in your computer array. As an example, M will be chosen to be 100. Remember, M must be an even number. The first point in the filter kernel is in array location 0, while the last point is in array location 100. This means that the entire signal is 101 points long. The center of symmetry is at point 50, i.e., $M/2$. The 50 points to the left of point 50 are symmetrical with the 50 points to the right. Point 0 is the same value as point 100, and point 49 is the same as point 51. If you must have a specific number of samples in the filter kernel, such as to use the FFT, simply add zeros to one end or the other. For example, with $M = 100$, you could make samples 101 through 127 equal to zero, resulting in a filter kernel 128 points long.